9.2.7 Applications of limits

Example 9.25

Alcohol is removed from the body by the lungs, the kidneys, and by chemical processes in liver. At moderate concentration levels, the majority work of removing the alcohol is done by the liver; less than 5% of the alcohol is eliminated by the lungs and kidneys. The rate r at which the liver processes alcohol from the bloodstream is related to the blood alcohol concentration x by a rational function of the form $r(x) = \dfrac{\alpha x}{x+\beta}$ for some positive constants α and β. Find the maximum possible rate of removal.

Solution

As the alcohol concentration x increases the rate of removal increases.

Therefore, the maximum possible rate of removal $= \lim\limits_{x\to\infty} r(x)$

$$= \lim_{x\to\infty} \frac{\alpha x}{x+\beta} = \lim_{x\to\infty} \frac{\alpha}{\left(1+\dfrac{\beta}{x}\right)} = \alpha.$$

9.2.8 Sandwich Theorem

Sandwich theorem is also known as squeeze theorem. As shown in the figure 9.27, if $f(x)$ is 'squeezed' or 'sandwiched' between $g(x)$ and $h(x)$ for all x close to x_0, and if we know that the functions g and h have a common limit l as $x \to x_0$, it stands to reason that f also approaches l as $x \to x_0$.

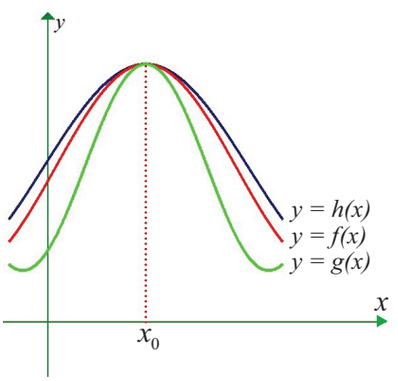


Fig. 9.27

Theorem 9.5 (Sandwich Theorem)

If $f, g, h : I \subseteq \mathbb{R} \to \mathbb{R}$ such that $g(x) \le f(x) \le h(x)$ for all x in a deleted neighbourhood of x_0 contained in I, and if

$$\lim_{x\to x_0} g(x) = \lim_{x\to x_0} h(x) = l, \quad \text{then} \quad \lim_{x\to x_0} f(x) = l.$$

Example 9.26

According to Einstein's theory of relativity, the mass m of a body moving with velocity v is $m = \dfrac{m_0}{\sqrt{1-\dfrac{v^2}{c^2}}}$, where m_0 is the initial mass and c is the speed of light. What happens to m as $v \to c^-$. Why is a left hand limit necessary?

Solution

$$\lim_{v\to c^-}(m) = \lim_{v\to c^-} \frac{m_0}{\sqrt{1-\dfrac{v^2}{c^2}}} = \frac{m_0}{\sqrt{\lim\limits_{v\to c^-}\left(1-\dfrac{v^2}{c^2}\right)}}$$

For $h > 0$, $c - h < v < c$. This implies, $(c-h)^2 < v^2 < c^2$.

That is, $\dfrac{(c-h)^2}{c^2} < \dfrac{v^2}{c^2} < 1$. That is, $\lim\limits_{h\to 0} \dfrac{(c-h)^2}{c^2} < \lim\limits_{h\to 0} \dfrac{v^2}{c^2} < \lim\limits_{h\to 0} 1$.

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